

JUSTIN KIM

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Education

University of Wisconsin–Madison | GPA: 3.76

B.S. in Computer Science & Data Science

Expected Graduation: Dec 2026

Madison, WI

Georgia Institute of Technology

M.S. in Computer Science, Machine Learning Specialization

Incoming Spring 2027

Online, Part-Time

Work Experience

Capital One

Jun 2026 – Aug 2026

Software Engineer Intern

Chicago, IL

- Built an AI-assisted authoring system that generates credit card fulfillment workflow configs from plain-English requirements, cutting a 2+ day manual process to under an hour.
- Designed a GraphRAG pipeline that grounded LLM output in approved workflow actions and validated each config against its schema, reaching 100% structural validity and 88% required-field completeness across 70 generated configs.
- Implemented real-time graph validation in a React/Monaco config editor, flagging cycles, unreachable states, and invalid references before save, adopted by 20+ intent engineers.

MadLease

Apr 2025 – Present

Co-Founder & Founding Engineer

Madison, WI

- Founded and grew a UW-Madison sublease marketplace to 250+ accounts, 100+ listings, and 5+ building partners.
- Replaced boost/recency-only ordering with a PostgreSQL ranking system across rent, campus proximity, and 8+ filters while preserving feed variety across sessions.
- Eliminated backend image byte handling with browser-direct S3 uploads via presigned URLs, adding retry and rollback logic to prevent incomplete listings on upload failure.

Integral

May 2025 – Aug 2025

Full Stack Software Engineer Intern

Madison, WI

- Replaced hard-coded crash-intake logic with configurable per-workflow documentation rules, enabling providers to define claim requirements through an Angular/Spring Boot platform.
- Engineered an OCR module in Angular/Tesseract.js that polled camera frames in a background worker and re-read VIN, plate, and odometer until each passed format and lookup validation, reaching 90%+ accuracy and company-wide reuse.

Denari

Feb 2025 – May 2025

Software Engineer, Capstone Project

Madison, WI

- Developed a citation-grounded RAG pipeline (OpenAI embeddings, pgvector/TimescaleDB, S3) over 50,000+ tax documents, scoring 61% on 2,200+ CPA-exam questions and outperforming a GPT-4o-mini baseline by 5% in accuracy

Astro AI Trading

Jan 2024 – Jan 2025

Machine Learning Engineer

Remote

- Led a 3-engineer team building per-ticker forecasting models across technical, macroeconomic, and news-sentiment features from 60+ daily sources, reducing held-out RMSE by 16% versus a price-only baseline.
- Fine-tuned a BERT 3-class news-sentiment classifier (positive/negative/neutral), reaching 77% test accuracy.

Projects

QuizClash | *Devpost* | *Source Code*

Node.js | **React** | **WebSocket** | **AWS**

- Created a real-time multiplayer quiz app converting PDFs and YouTube videos into Kahoot-style games with synchronized WebSocket gameplay and public/private rooms.
- Automated quiz creation with an ML pipeline for transcription, content extraction, question generation, and distractor selection using OCR, Whisper, T5, RoBERTa, and Sense2Vec, reaching 75% manually evaluated relevance.

Technical Skills

Languages: Python, Java, JavaScript/TypeScript, Go, C, SQL, HTML/CSS

Libraries/Frameworks: React, Spring Boot, Node.js/Express, Flask, PyTorch, Angular, Scikit-Learn

Tools: Git, Docker, AWS (S3, EC2, Lambda, CloudFront, IAM, CDK), PostgreSQL, TimescaleDB/pgvector, Firebase